# Occupational

Community Medicine (PSM) | 3rd Year

Hi Sammy!
Occupational diseases
= guaranteed Qs!
Learn the classics!

Occupational Health | PSM | 3rd Year | MBBS

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Classification of hazards is how examiners start every LAQ on this topic!

OCCUPATIONAL DISEASE

*A disease arising out of or in the course of employment — directly related to workplace exposure.*

## Classification of Occupational Hazards

| Type | Agent / Factor | Examples of Disease |
|---|---|---|
| **Physical** | Heat, cold, noise, radiation | Heat stroke, NIHL, radiation sickness |
| **Chemical** | Dust, fumes, gases, metals | Pneumoconiosis, lead/Hg poisoning |
| **Biological** | Bacteria, virus, fungi | Anthrax, brucellosis, leptospirosis |
| **Mechanical** | Posture, repetition, vibration | Carpal tunnel, back pain, HAVS |
| **Psychosocial** | Stress, shift work, harassment | Burnout, anxiety, depression |

## ILO Classification of Occupational Diseases

| Category | Examples |
|---|---|
| **Diseases by agents (chemical)** | Lead, mercury, arsenic, benzene poisoning |
| **Diseases by agents (physical)** | NIHL, radiation injury, vibration syndrome |
| **Diseases by biological agents** | Anthrax, brucellosis, hepatitis B/C |
| **Diseases by target organ** | Respiratory (asbestosis), skin (dermatitis) |
| **Occupational cancer** | Mesothelioma, bladder Ca (dyes), scrotal Ca (soot) |

## Hierarchy of Prevention (Most to Least Effective)

1. **Elimination** — Remove the hazard entirely from workplace
2. **Substitution** — Replace hazardous substance with safer one
3. **Engineering** — Ventilation, enclosure, isolation of process
4. **Administrative** — Job rotation, training, reduced exposure time
5. **PPE** — Masks, gloves, goggles — LAST resort (least effective)

Silicosis vs Asbestosis comparison table = THE most repeated LAQ ever!

## Pneumoconioses — THE Comparison Table

| Feature | Silicosis | Asbestosis | Byssinosis |
|---|---|---|---|
| **Dust** | Free silite (SiO2) | Asbestos fibres | Cotton dust |
| **Occupation** | Mining, sandblasting | Insulation, shipyard | Textile (cotton mills) |
| **X-ray** | Egg-shell calcific. | Pleural plaques | Normal early |
| **Complication** | TB (silicotuberculosis) | Mesothelioma, lung Ca | Chronic bronchitis |
| **Pathology** | Silicotic nodules | Asbestos bodies | Byssinosis grades |
| **Prognosis** | Progressive, no Rx | Progressive, cancer | Reversible if early |
| **Special** | Most common P-cosis | Most fibrogenic | Monday fever |

★ **Silicosis = Most COMMON | Asbestosis = Most FIBROGENIC**
Byssinosis Monday fever = weekend off, symptoms return on Monday with cotton dust!

## Lead Poisoning (Plumbism) — Complete Picture

| Aspect | Details |
|---|---|
| **Sources** | Battery industry, paint, smelting, printing, cosmetics (surma/kajal) |
| **Absorption** | Inhalation (main industrial), ingestion (children — pica) |
| **Blood level** | >10 ug/dL (children), >40 ug/dL (adults) = toxic |
| **Hematological** | Basophilic stippling, sideroblastic anaemia, raised ALA in urine |
| **GI** | Lead colic (severe abd pain), constipation, Burton's line (blue gum line) |
| **Neuro** | Wrist drop (adults), encephalopathy (children), foot drop |
| **Renal** | Chronic nephritis, Fanconi syndrome |
| **Diagnosis** | Blood lead level, urine ALA, FEP, basophilic stippling on PBS |
| **Treatment** | Chelation: EDTA (adults), DMSA/Succimer (children, oral) |

★ **Lead ABCDs: Anaemia, Burton's line, Colic, Drop (wrist/foot)**
Basophilic stippling = PATHOGNOMONIC finding on peripheral blood smear!

NIHL audiogram pattern and TLV concepts are asked in viva often!

## Noise-Induced Hearing Loss (NIHL)

| | |
|---|---|
| Permissible limit: | 85 dB for 8 hrs/day (OSHA) — TLV |
| Doubling rule: | Every 3 dB increase = halve exposure time |
| Audiogram: | 4 kHz DIP (notch) = earliest and characteristic finding |
| Type: | Sensorineural hearing loss (irreversible) |
| Prevention: | Engineering controls > ear plugs > ear muffs |
| Temporary Threshold Shift: | Recovers after rest; permanent if continued |

## Heat-Related Occupational Illness

| Condition | Mechanism | Features / Treatment |
|---|---|---|
| **Heat cramps** | Salt loss from sweating | Muscle cramps — Rx: oral salt solution |
| **Heat exhaustion** | Water + salt depletion | Weakness, tachycardia — Rx: IV fluids |
| **Heat stroke** | Thermoregulatory failure | Temp >40.6C, NO sweating — EMERGENCY |
| **Heat syncope** | Peripheral vasodilation | Fainting — Rx: rest, fluids, shade |

## Other Important Occupational Diseases

| Disease | Agent / Cause | Key Feature |
|---|---|---|
| **Mercury poisoning** | Thermometer, dental | Erethism, tremors, Minamata disease |
| **Chrome ulcers** | Chromium compounds | Nasal septum perforation, lung Ca |
| **Bagassosis** | Sugarcane dust | Hypersensitivity pneumonitis |
| **Farmer's lung** | Mouldy hay (thermophilic) | Type III + IV hypersensitivity |
| **Anthracosis** | Coal dust | Coal worker's pneumoconiosis |
| **Berylliosis** | Beryllium | Granulomas (sarcoid-like), lung fibrosis |
| **Mesothelioma** | Asbestos | Malignant pleural tumour — pathognomonic |

★ **TLV (Threshold Limit Value) — ACGIH concept:**

TLV-TWA: 8-hr avg safe level | TLV-STEL: 15-min max | TLV-C: absolute ceiling

MAC (Maximum Allowable Concentration) = same concept, European term

# Legislation — Factories Act, ESI Act

Acts and laws are dry but easy marks! Just learn the key numbers!

## Factories Act, 1948 (Amended 1987)

| | |
|---|---|
| Definition: | Factory = 10+ workers with power OR 20+ without power |
| Working hrs: | Max 48 hrs/week, 9 hrs/day; overtime = double wages |
| Women: | Cannot work between 7 PM and 6 AM (night shift restriction) |
| Children: | <14 years = prohibited; 15-18 = adolescent (restricted work) |
| Health: | Cleanliness, ventilation, lighting, drinking water, latrines |
| Safety: | Fencing of machinery, fire escapes, first aid boxes |
| Welfare: | Canteen (>250 workers), creche (>30 women), rest rooms |

## ESI Act, 1948 (Employees' State Insurance)

| | |
|---|---|
| Coverage: | Factories with 10+ employees; wage limit Rs 21,000/month |
| Contribution: | Employer 3.25% + Employee 0.75% of wages |
| Benefits: | 6 types — medical, sickness, maternity, disability, dependant, funeral |
| Medical benefit: | Free treatment for worker + family at ESI hospitals/dispensaries |
| Sickness benefit: | 70% of wages for max 91 days in a year |
| Maternity: | Full wages for 26 weeks (as per Maternity Benefit Act 2017) |

## Other Relevant Labour Laws

| Act | Key Provision |
|---|---|
| Workmen's Compensation 1923 | Employer pays for injury/death during employment |
| Mines Act 1952 | Safety in mines; min age 18; rescue stations mandatory |
| Child Labour Act 1986 | Bans child (<14) labour in hazardous occupations |
| BOCW Act 1996 | Building & construction workers welfare + safety |
| Occupational Safety Code 2020 | New code merging 13 old labour laws into one |

# Exam Strategy + High Yield

Occupational health =
scoring goldmine in PSM!
You're doing great!

## Most Repeated LAQs (10–12 marks)

1. Classify occupational hazards. Describe pneumoconioses in detail
2. Compare silicosis vs asbestosis vs byssinosis (table format)
3. Describe lead poisoning — sources, clinical features, diagnosis, treatment
4. Write about Factories Act 1948 — health, safety, welfare provisions
5. Describe ESI Act — coverage, contributions, benefits (6 types)

## Most Repeated SAQs (5–6 marks)

1. Noise-induced hearing loss — features and audiogram
2. Lead poisoning — clinical features (ABCD mnemonic)
3. Heat stroke vs heat exhaustion
4. Byssinosis — Monday fever, grades, prevention
5. TLV — types and significance
6. Hierarchy of prevention in occupational health

## One-Liner Associations (MCQ/Viva Gold)

| Question | Answer |
|---|---|
| Most common pneumoconiosis | Silicosis |
| Most fibrogenic pneumoconiosis | Asbestosis |
| Monday fever / Monday feeling | Byssinosis (cotton dust) |
| Egg-shell calcification on X-ray | Silicosis (hilar LN) |
| Pathognomonic for asbestos | Mesothelioma / Asbestos bodies |
| Burton's line | Lead poisoning (blue line on gums) |
| Wrist drop — occupational | Lead poisoning |
| 4 kHz audiometric dip | NIHL (earliest sign) |
| Permissible noise level | 85 dB for 8 hours (OSHA) |
| Factory definition | 10+ with power / 20+ without |
| ESI employee contribution | 0.75% of wages |
| Erethism + tremors | Mercury poisoning |

♥ **Protecting workers is protecting families, Sammy!**
You'll make workplaces safer someday. Learn this well!